VERIDIAN PHARMA LTD. — Quality & Compliance Directorate, Hyderabad Facility

Standard Operating Procedure: Cleaning Validation and Surface Monitoring

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Owner: Meera Rao (QA Head)	**Department:** Quality Assurance	**Status:** Effective

1. Purpose

To establish the scientific framework, acceptance limits, sampling methodologies, and revalidation frequencies for validating cleaning procedures applied to shared pharmaceutical manufacturing equipment at Veridian Pharma Ltd., Hyderabad.

Thorough cleaning validation prevents cross-contamination between consecutive product campaigns, eliminates residual detergents, and controls microbial bioburden to ensure patient safety.

Complies with cGMP rules regarding dedicated vs non-dedicated facility equipment controls.

2. Scope

Applies to all product-contact processing equipment utilized in formulation, compounding, blending, granulation, compression, and primary packaging operations.

Governs both automated clean-in-place (CIP) and manual clean-out-of-place (COP) procedures.

Production operators, validation engineers, QC analytical testing staff, and QA auditors must adhere to this document.

3. Responsibilities

• **Validation Specialist:** Draft validation protocols, calculate maximum allowable carryover (MACO), execute swab sampling, and prepare validation reports.

• **Quality Control Analyst:** Analyze swab and rinse recovery samples using validated HPLC, TOC, and microbial testing methods.

• **QA Head (Meera Rao):** Approve cleaning validation protocols, evaluate residue limits, and authorize formal validation reports.

4. Procedure

4.1 Scientific Cleaning Rationale: Cleaning procedures must be validated to confirm the effective removal of previous product residues, cleaning agents, and microbial bioburden from equipment contact surfaces.

4.2 Worst-Case Product Matrix: Worst-case products and equipment grouping matrices must be evaluated based on solubility, toxicity, and cleanability to determine representative challenging products.

4.3 Validation Execution Runs: Three consecutive successful runs required.

4.4 Swab Analytical Limit: Swab residue limit 10 ppm (fictional).

4.5 Visual Cleanliness Verification: Visual inspection must confirm the equipment surface is clean, dry, and free of visible particulate or discoloration under 500 lux illumination.

4.6 Rinse Water Analytical Verification: Rinse samples must be analyzed for total organic carbon (TOC) and conductivity where appropriate to verify automated CIP cycles.

4.7 Microbiological Bioburden Compliance: Microbiological swab testing must meet specified colony-forming unit (CFU) limits established in the validation protocol.

4.8 Revalidation Frequency: Revalidation every 24 months.

4.9 Change Assessment Mandate: Any modification to cleaning agent formulation, critical cleaning parameters, or equipment surface requires a formal change assessment before adoption.

5. Records and Documentation

Cleaning validation protocols, swab sampling recovery tests, chromatographic analytical raw logs, and validation summary reports must be stored in the Validation Directorate archives.

6. Revision History

Version	Effective Date	Change Description
1.0	2026-06-18	Initial release of cleaning validation master procedure.